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OMM Lab Prep Lecture: Diagnosis of the Elbow and Wrist

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Session Objectives

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1. Know relevant dermatomes, myotomes and peripheral nerve distributions
2. Review relevant anatomy of the elbow, forearm and wrist
3. Diagnose somatic dysfunction of the ulnohumeral joint
4. Diagnose somatic dysfunction of the radial head (radioulnar joint)
5. Diagnose somatic dysfunction of the radiocarpal joint
6. Understand the relationship between the ulnohumeral and radiocarpal dysfunction

Dermatomes and Myotomes

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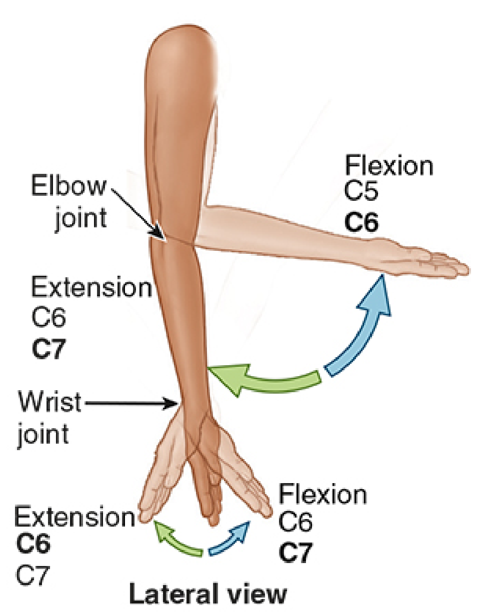
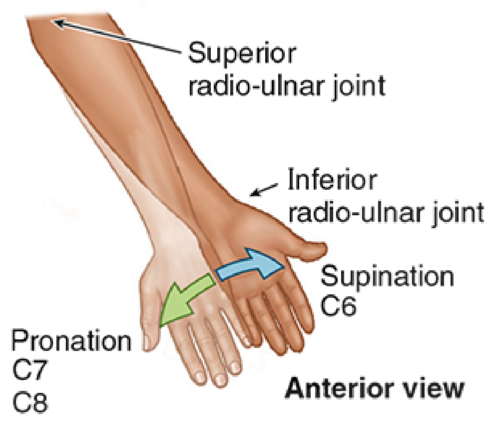
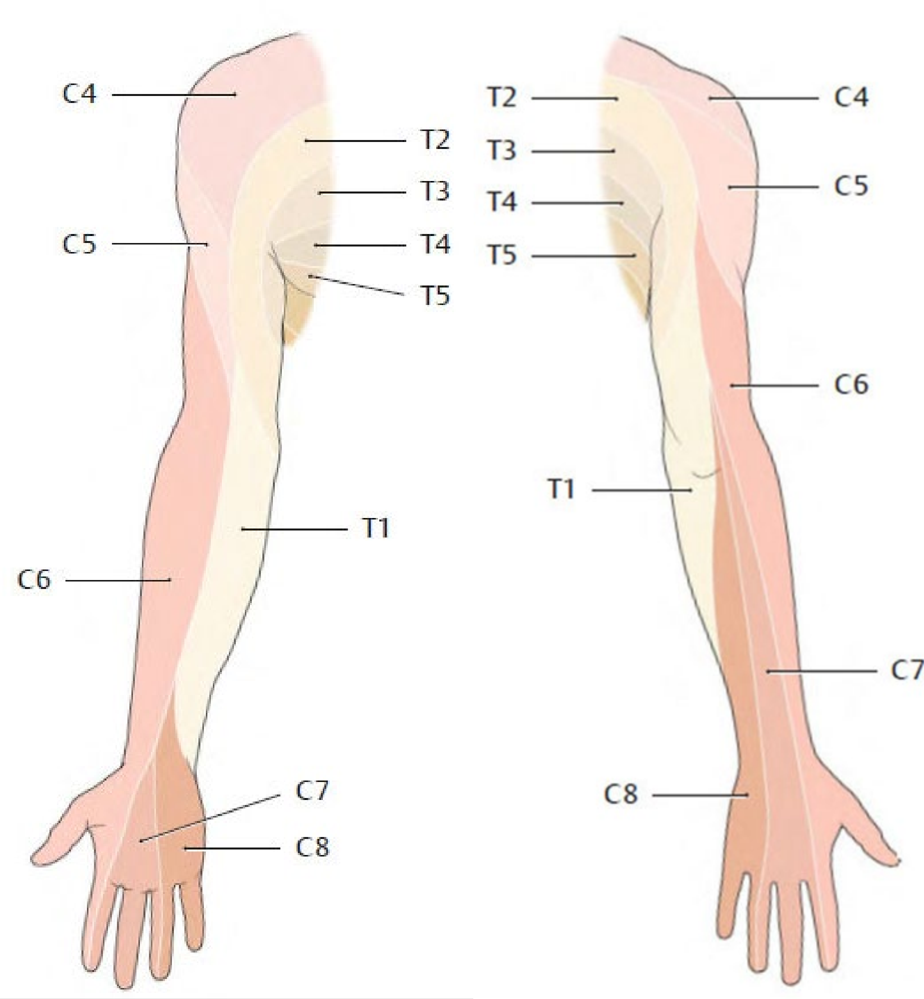
	DERMATOME TESTING POINT	MYOTOME
C5	lateral antecubital fossa just proximal to elbow joint	elbow flexion – biceps and brachialis biceps reflex
C6	dorsal proximal phalanx of thumb	wrist extension – extensor carpi radialis longus and brevis biceps reflex supinator reflex
C7	dorsal proximal phalanx of middle finger	elbow extension – triceps triceps reflex
C8	dorsal proximal phalanx of little finger	middle finger flexion – flexor digitorum profundus (distal phalanx)
T1	medial antecubital fossa just proximal to elbow joint	little finger abduction – abductor digiti minimi, interossei

Watson·Anatomy, Laura Jayne. "Nerve Supply to the Upper Limb." *Geeky Medics*, 13 Nov. 2015, https://geekymedics.com/nerve-supply-to-the-upper-limb/

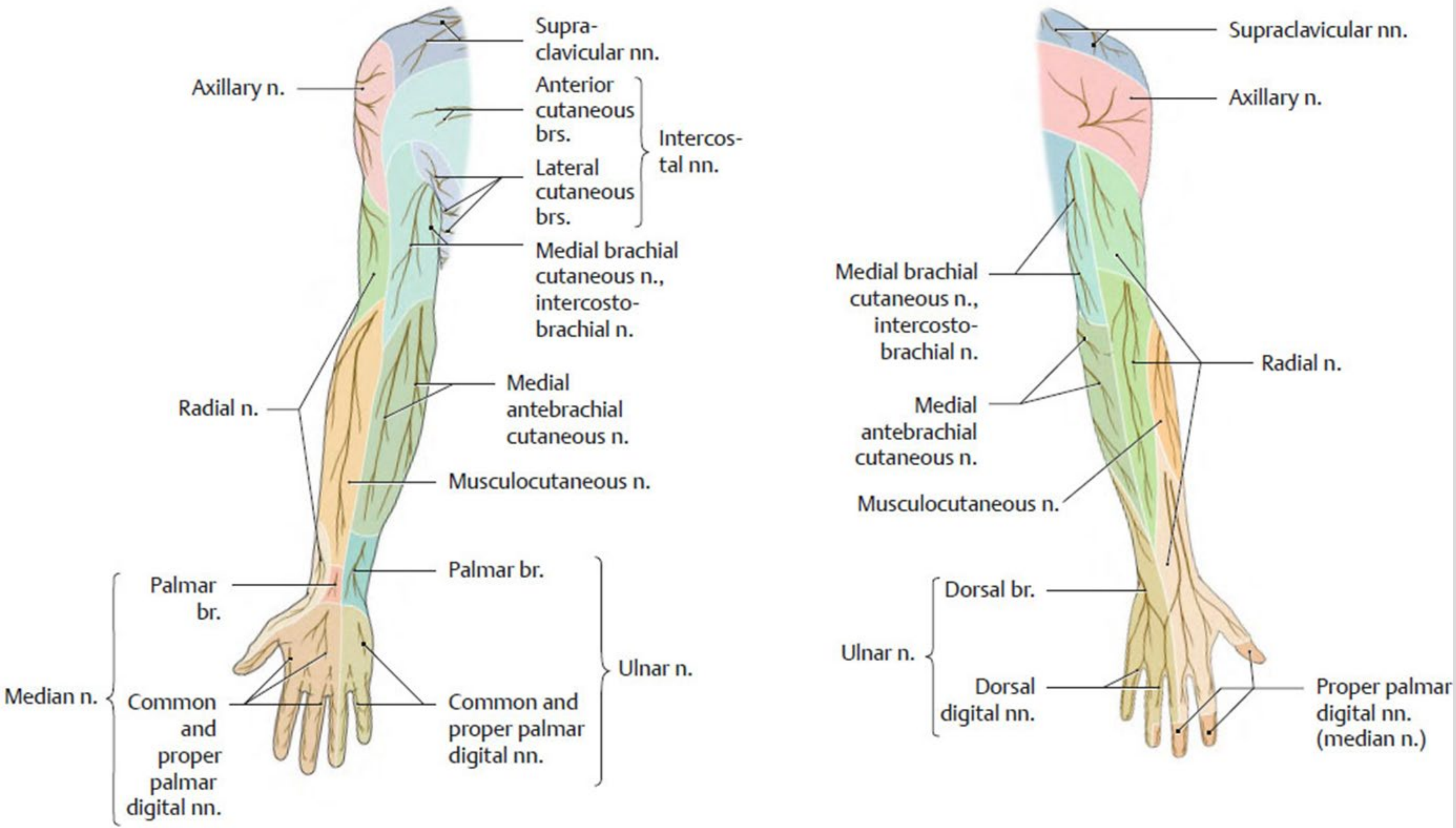






Dalley, A. F., II, & Agur, A. M. R. (2023). *Moore's clinically oriented anatomy, 9e*. Lippincott Williams & Wilkins, a Wolters Kluwer business.

Gilroy, A. M., MacPherson, B. R., & Wikenheiser, J. C. (2020). *Atlas of Anatomy* (4th ed.). Thieme Medical Publishers Inc..



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A Anterior view.

B Posterior view.

Elbow Joint

Consists of 3 synovial joints

- Ulnohumeral
  - Primary motions: flexion and extension (hinge joint)
  - Secondary motions: ABduction and ADduction
- Radiohumeral
  - Primary motions: pronation and supination
- Radioulnar
  - Primary motions: pronation and supination

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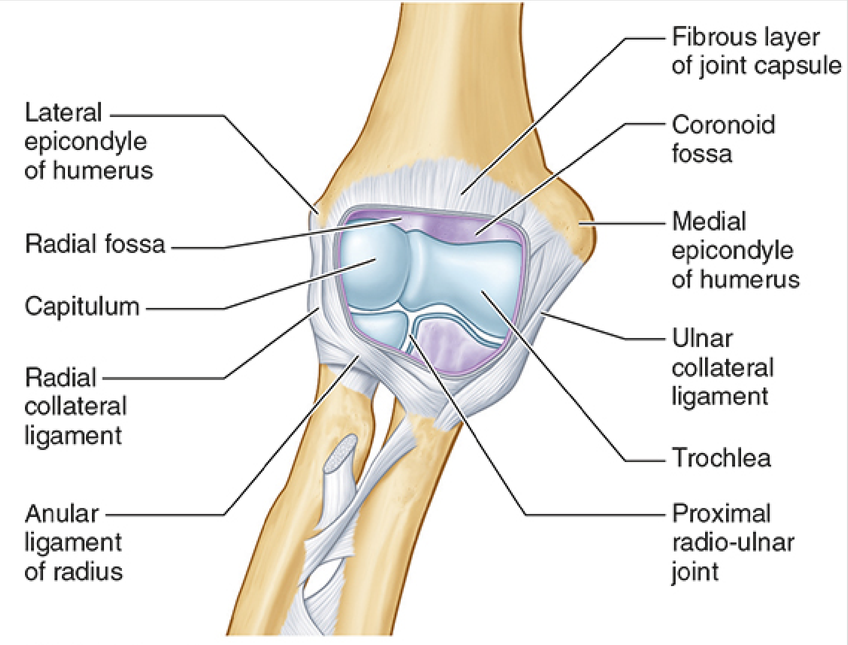

(A) Anterior view

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Elbow Joint

- Radial (lateral) collateral ligament (RCL)
  - Lateral epicondyle → annular ligament of radial head and radial notch
- Ulnar (medial) collateral ligament (UCL)
  - Medial epicondyle → coronoid and olecranon processes

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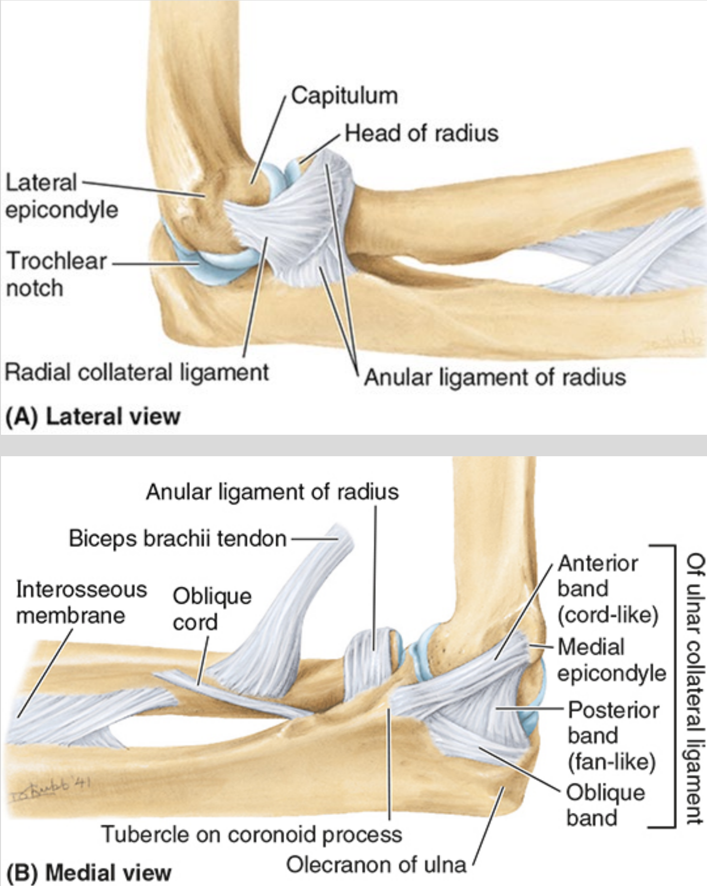

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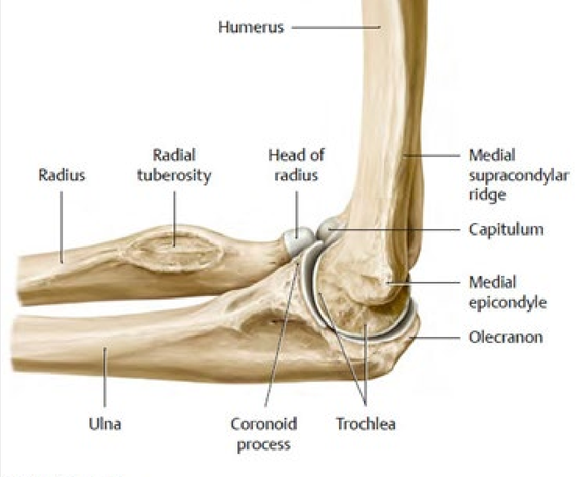
Ulnohumeral Joint

- Elbow flexion is anatomically limited by the coronoid fossa and coronoid process
- Elbow extension is anatomically limited by the olecranon fossa and olecranon process
- The looseness between the trochlea and trochlear notch allows the:
  - Medial ulnar glide with distal forearm ABduction and supination
  - Lateral ulnar glide with distal forearm ADduction and pronation



C Medial view.

A Anterior view.
B Posterior view.

Gilroy, A. M., MacPherson, B. R., & Wikenheiser, J. C. (2020). *Atlas of Anato*

Ulnohumeral Joint

- Spiraled fit of trochlea and trochlear notch is responsible for the carrying angle when the elbow is extended and supinated

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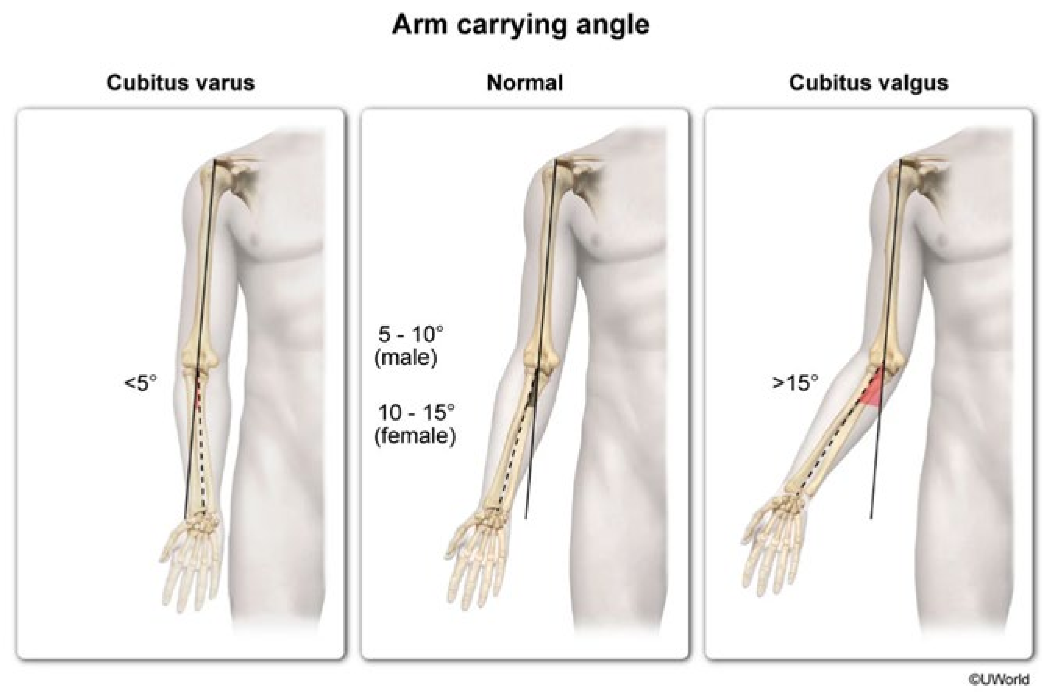

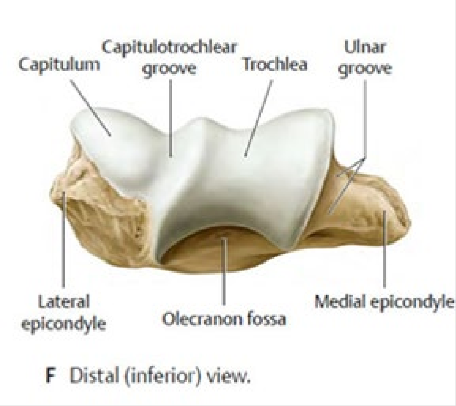

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Ulnohumeral Joint Somatic Dysfunction

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- Flexion SD and Extension SD is primarily due to a hypertonic biceps brachii and triceps brachii respectively
- ABduction SD and ADduction SD is primarily a preference of medial or lateral glide of the proximal ulna respectively
  - Medial and lateral glide are accessory rocking motions of the proximal ulna on the trochlea of the humerus and can also be induced by supination and pronation respectively.

Diagnosis	Motion Preference
ABduction SD	Proximal ulna glides medially easily with distal forearm ABduction and does not glide laterally easily with distal forearm ADduction
ADduction SD	Proximal ulna glides laterally easily with distal forearm ADduction and does not glide medially easily with distal forearm ABduction
Flexion SD	Elbow flexes easily and does not extend easily
Extension SD	Elbow extends easily and does not flex easily

Ulnohumeral Joint Somatic Dysfunction

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Screening:

- Observation of carrying angle bilaterally.
- Active elbow flexion and extension range of motion bilaterally.

Diagnosis:

- Patient is seated. The physician stands in front of the patient.
- Physician grasps the proximal radioulnar area with both hands.
- Physician places the patient's wrist and hand between the physician's trunk and the medial side of the physician's elbow, to support it. Patient's elbow should be slightly flexed (5-45 degrees).
    - Physician can also cup the patient's elbow with one hand and the other hand grasps the patient's wrist or hand.
- Physician introduces distal forearm ABduction → proximal ulna moves medially while distal ulna moves laterally.
- Physician introduces distal forearm ADduction → proximal ulna moves laterally and distal ulna moves medially.

Figure 39.28: Evaluation of elbow abduction and adduction
Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 5e, 2026. (Photo courtesy of Western University of Health Sciences, Pomona, CA, with permission.)

Osteopathic Clinical Skills. "Somatic Dysfunction: Upper Extremity - Elbow (Ulna, Radial Head, Forearm Interosseous)." *YouTube*, 25 Feb. 2020, www.youtube.com/watch?v=_RXIT32GjvI. Accessed 26 Jan. 2026.

Radial Head

- The radiohumeral joint = radial head fovea and capitulum
- The radioulnar joint = radial head and radial notch
- Annular ligament is shaped like a tapered cup with the bottom broken out which prevents distal displacement of the radius

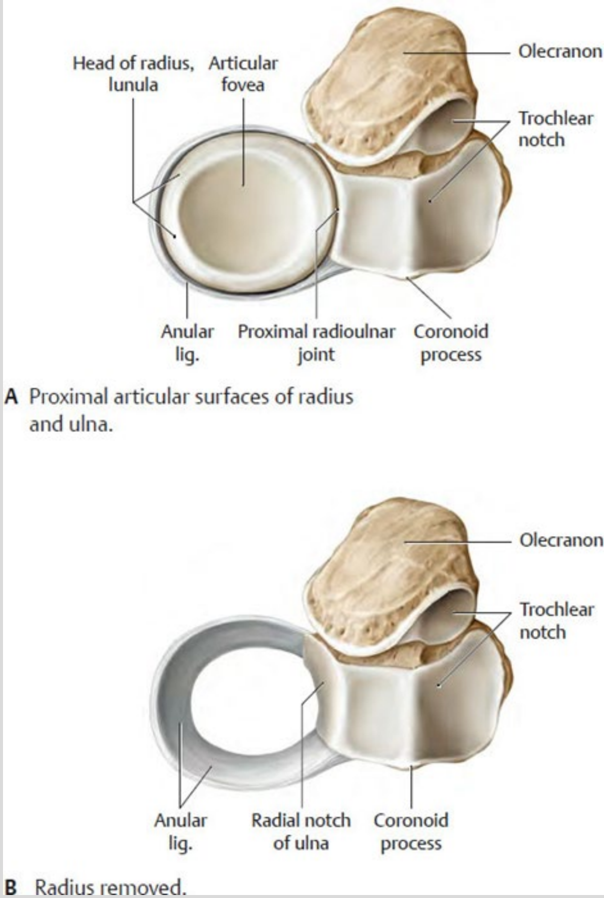

Fig. 26.8 Proximal radioulnar joint
Right elbow, proximal (superior) view.

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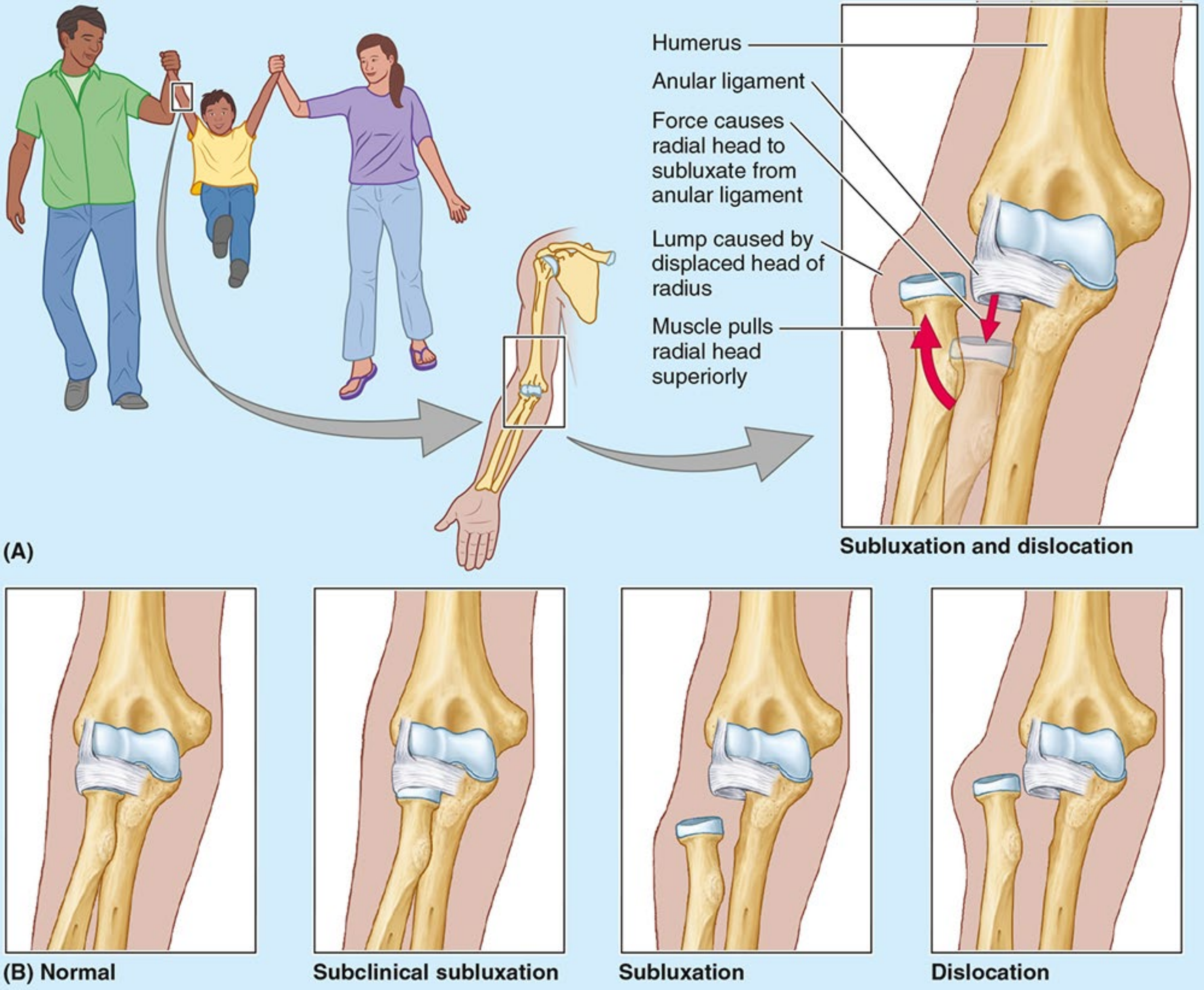
Clinical Connection – Nursemaid's Elbow

- Caused by sudden pull on extended and pronated arm
- Child will hold arm close with elbow flexed and forearm pronated
- Reduce with supination-flexion technique



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Radial Head Somatic Dysfunction

- Radial head motion is a combination of humeroradial and proximal radioulnar joint motion
- Radioulnar joint somatic dysfunction is primarily a preference of anterior and posterior glide of the radial head

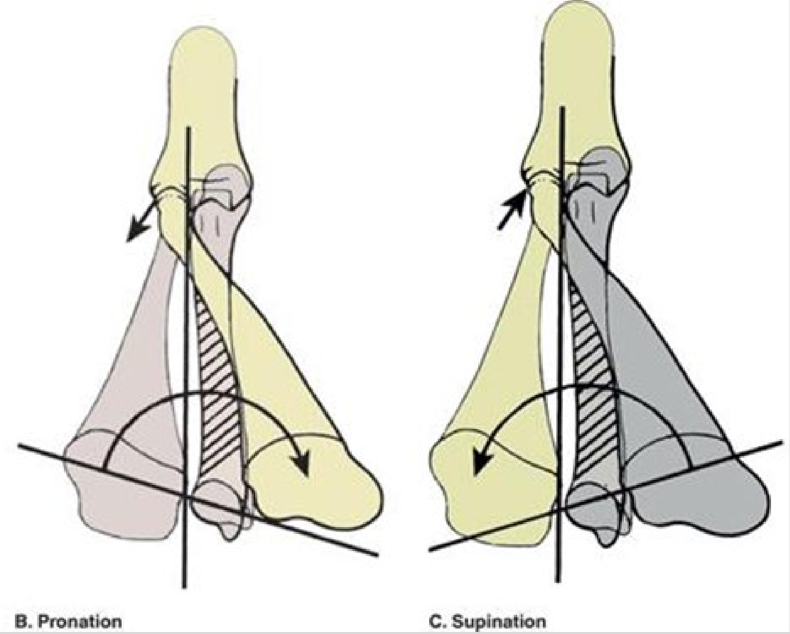

Figure 39.4 Forearm pronation and supination
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Radial Head Diagnosis	Motion Preference
Posterior Radial Head SD	Glides posteriorly easily with pronation and does not glide anteriorly easily with supination
Anterior Radial Head SD	Glides anteriorly easily with supination and does not glide posteriorly easily with pronation

Radial Head
Somatic Dysfunction

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Screening:

- Active forearm supination and pronation range of motion bilaterally.

Diagnosis:

- Patient is seated with arms in neutral position (no humoral flexion or extension) and elbow flexed to 90 degrees. The physician stands in front of the patient.
- Physician stabilizes the proximal radial head with one hand and grasps the patient's distal forearm with the other hand, placing it in a neutral position with the patient's thumb in a vertical position, pointed superiorly.
- Physician introduces pronation → distal radius moves anteriorly while the proximal radial head glides posteriorly.
    - Can also directly assess the radial head's ability to glide posteriorly independent of pronation.
- Physician introduces supination → distal radius moves posteriorly while the proximal radial head glides anteriorly.
    - Can also directly assess the radial head's ability to glide anteriorly independent of supination.
- Please note there can be restriction of forearm pronation or supination without restriction of radial head glide.

Figure 39.31 Evaluation of forearm pronation and supination and radial head motion
Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 5e, 2026. (Photo courtesy of Western University of Health Sciences, Pomona, CA, with permission.)

Radial Head Somatic Dysfunction

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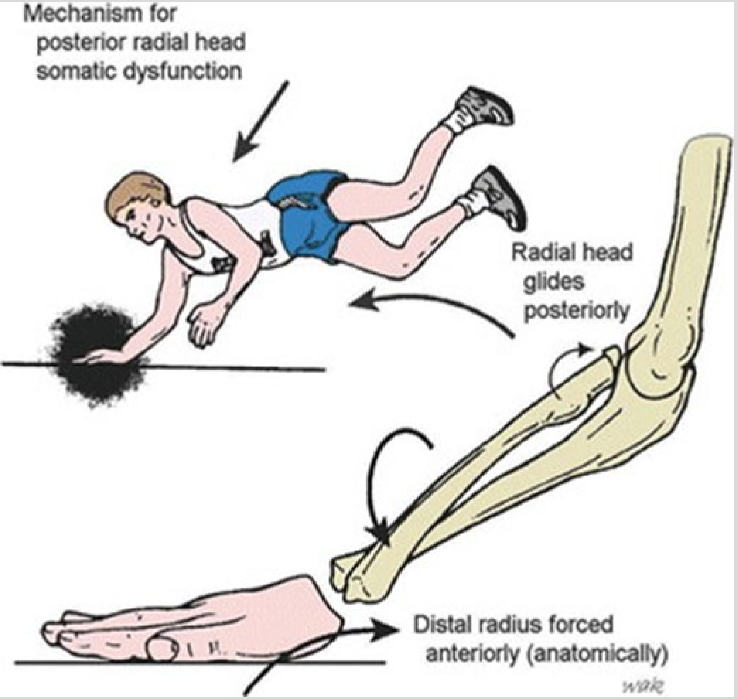


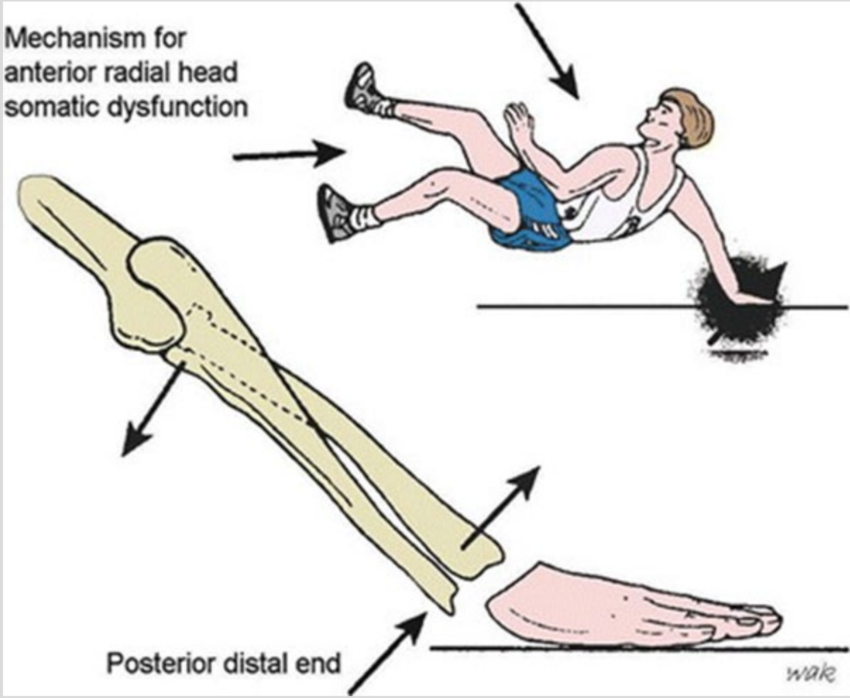
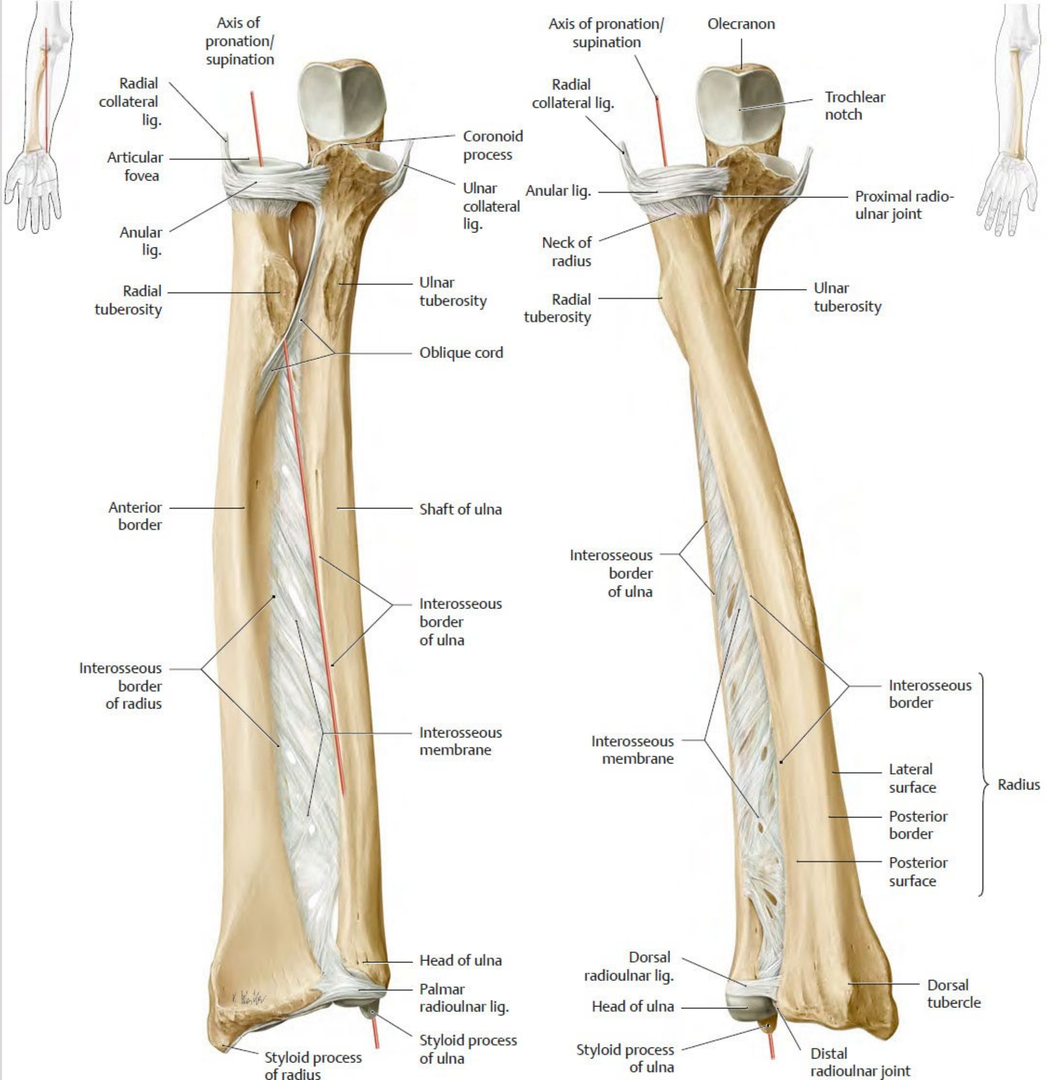


Figure 49C.1 Forward fall on outstretched hand and Figure 49C.2 Fall backward on extended arm
Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e, 2018. (Illustration by William A. Kuchera, DO, FAAO, with permission.)

Interosseous Membrane

- Majority of the fibers are orientated inferior and medial from the radius to the ulna
- This allows forces transmitted by the distal radiocarpal joint to transfer to the ulna and subsequently the humerus



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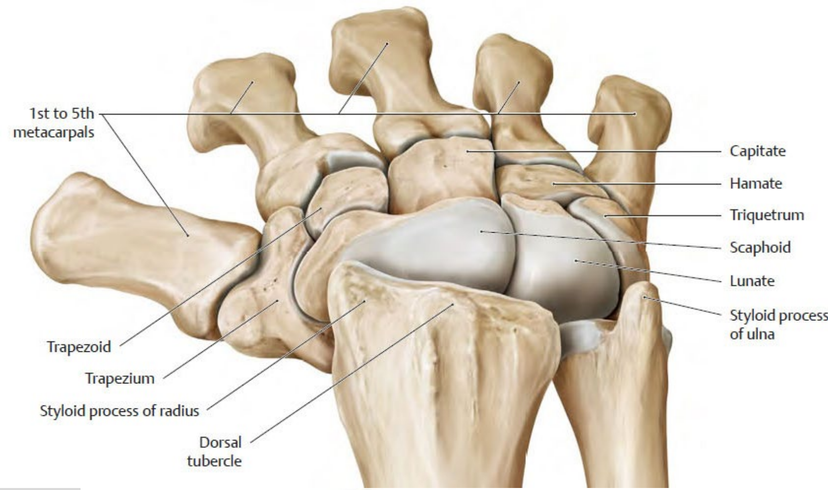
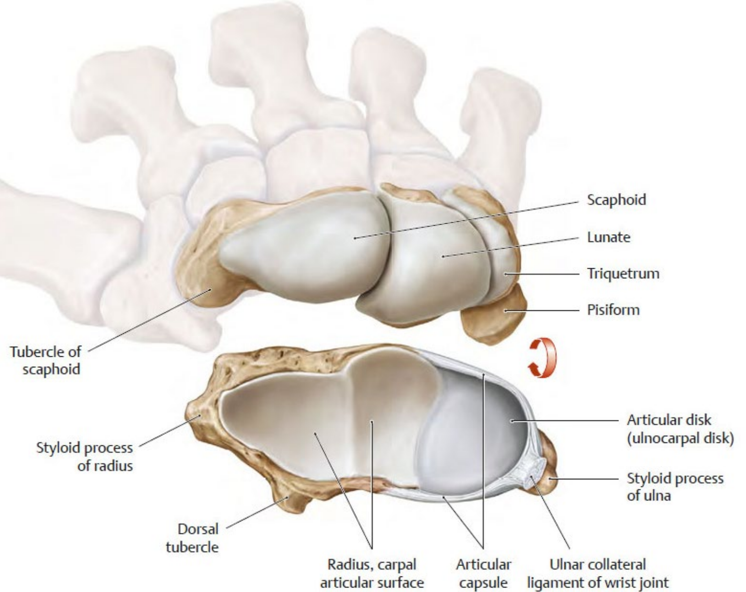
Wrist Joint

- Does not technically include the ulna
- Distal radius and radioulnar disk articulate with the proximal carpals, scaphoid, lunate, triquetrum
- Radiocarpal joint is a condyloid synovial joint





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Wrist Joint

- The wrist is stabilized anteriorly and posteriorly by palmar and dorsal radiocarpal ligaments
- Ulnar collateral ligament
  - Ulnar styloid process → triquetrum
- Radial collateral ligament
  - Radial styloid process → scaphoid

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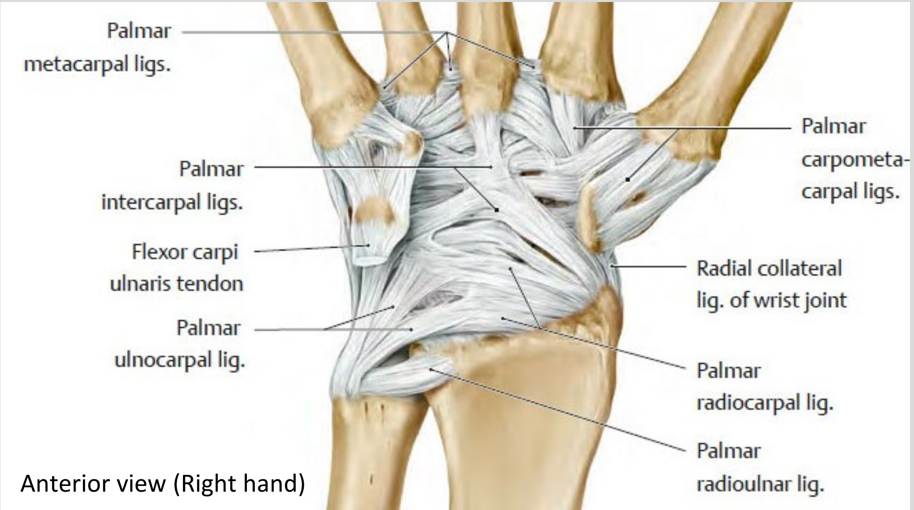
Posterior view (Right hand)



Anterior view (Right hand)

Distal Radioulnar Joint

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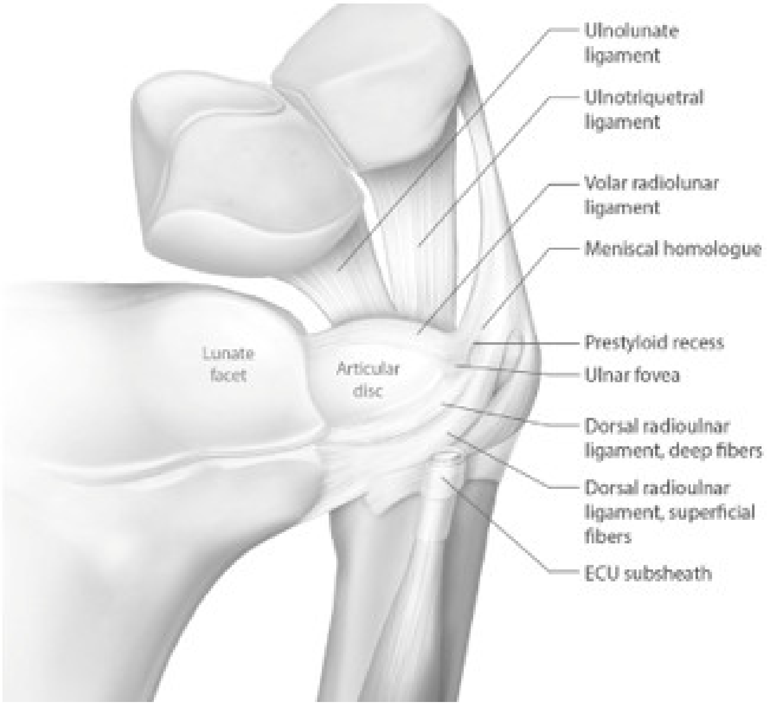
- Distal radioulnar joint is stabilized by the triangular fibrocartilage complex (TFCC)
  - Articular disk
  - Dorsal Radioulnar ligament
  - Ventral Radioulnar ligament
  - Dorsal Ulnocarpal ligament
  - Ventral Ulnocarpal ligament
  - Floor of extensor carpi ulnaris tendon sheath



Distal radioulnar joint. Right wrist, posterior (dorsal) view.

https://wikism.org/Triangular_Fibrocartilage_Complex

Radiocarpal Joint
Somatic Dysfunction

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- Somatic dysfunction of the wrist is not related to gross motions of the wrist but rather the small gliding motions of the proximal carpals
- Somatic dysfunction at the wrist is often a combination of anterior or posterior and medial or lateral glide somatic dysfunctions of the proximal carpals

Proximal Carpal Diagnosis	Motion Preference
Extension SD	Glides anteriorly easily with wrist extension and does not glide posteriorly easily with wrist flexion
Flexion SD	Glides posteriorly easily with wrist flexion and does not glide anteriorly easily with wrist extension
ADduction SD Ulnar Deviation SD	Glides laterally easily with wrist ulnar deviation or ADduction and does not glide medially easily with wrist radial deviation or ABduction
ABuction SD Radial Deviation SD	Glides medially easily with wrist radial deviation or ABduction and does not glide laterally easily with wrist ulnar deviation or ADduction

Radiocarpal Joint Somatic Dysfunction

Screening:

- Active wrist abduction, adduction, flexion, and extension bilaterally

Diagnosis:

- Patient is seated. Physician stands in front of the patient.
- Patient’s arms are at the sides, with the elbows flexed to 90 degrees.
- Physician tests the following motions in each of the patient’s wrists by stabilizing the patient’s wrist.
  - Please note ulnar and radial deviation restrictions may differ when the forearm is positioned in pronation or supination due to reciprocal motion of ulnohumeral and radiocarpal joints.
  - Restriction of wrist flexion and extension does not significantly differ with forearm supination or pronation.
- Physician introduces wrist extension → proximal carpals glide anteriorly.
- Physician introduces wrist flexion → proximal carpals glide posteriorly.
- Physician introduces wrist ulnar deviation or ADduction → proximal carpals glide laterally
- Physician introduces wrist radial deviation or ABduction → proximal carpals glide medially
- Please note physician can also directly assess individual carpal bone glides in order to diagnose scaphoid, lunate, or triquetrum SD.

Figure 39.32 Evaluation of wrist extension with forearm supinated
Figure 39.33 Evaluation of wrist flexion with forearm supinated
Figure 39.34 Evaluation of wrist ulnar deviation with forearm supinated
Figure 39.35 Evaluation of wrist radial deviation with forearm supinated

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Ulnohumeral and Radiocarpal Dysfunction

- Reciprocal motion is present between the elbow and wrist due to the parallelogram effect.
- With ulnohumeral ABduction SD **(A)** medial glide is preferred at the proximal ulna which likely translates to preferred lateral glide of the distal radius and radiocarpal ADduction SD.
- With ulnohumeral ADduction SD **(C)** lateral glide is preferred at the proximal ulna which likely translates to preferred medial glide of the distal radius and radiocarpal ABduction SD.

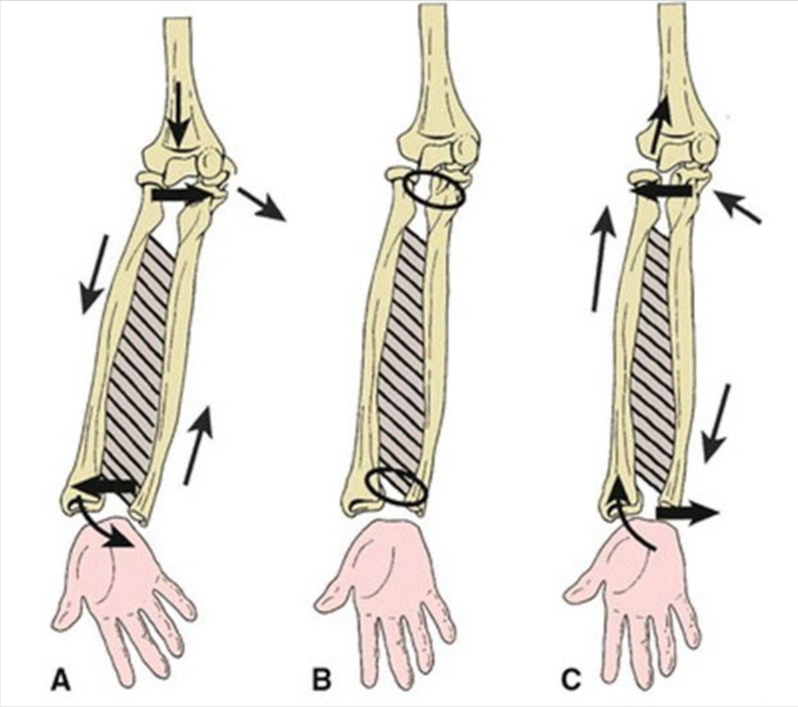

Figure 49C.4
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References

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1. Heinking, K. P., Giusti, R. E., Hensel, K. L., & Kuchera, W. A. (2026). F*oundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research*, *5e*. Lippincott Williams & Wilkins, a Wolters Kluwer business. https://nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=261514817&bookid=3456#261515052
2. Seffinger, M. A., Hruby, R., Willard, F. H., Licciardone, J., Kuchera, W. A., Rey, M., Tang, M. Y., Muhlestein, J., Lewis, A., Burton, A., Garispe, A., Guevarra, C., Koh-Pham, C., Rouse, N., Hecht, K., Helf, S., Marquez, A. P., & Seffinger, A. (2018). *Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research*, *4e*. Lippincott Williams & Wilkins, a Wolters Kluwer business. https://nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=209559343&bookid=2582
3. Dalley, A. F., II, & Agur, A. M. R. (2023). *Moore's clinically oriented anatomy, 9e*. Lippincott Williams & Wilkins, a Wolters Kluwer business.
4. DiGiovanna, E. L., Amen, C. J., & Burns, D. K. (2021). *An osteopathic approach to diagnosis and treatment, 4e*. Lippincott Williams & Wilkins, a Wolters Kluwer business. https://nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=248815018&bookid=2969
5. DiGiovanna, E. L., Amen, C. J., & Burns, D. K. (2021). *An osteopathic approach to diagnosis and treatment, 4e*. Lippincott Williams & Wilkins, a Wolters Kluwer business. https://nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=248815090&bookid=2969#248815121
6. Gilroy, A. M., MacPherson, B. R., & Wikenheiser, J. C. (2020). *Atlas of Anatomy* (4th ed.). Thieme Medical Publishers Inc..
7. Watson·Anatomy, Laura Jayne. "Nerve Supply to the Upper Limb." *Geeky Medics*, 13 Nov. 2015, https://geekymedics.com/nerve-supply-to-the-upper-limb/

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